maya

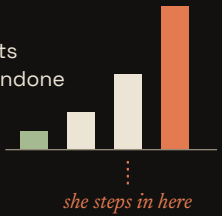
“He typed his own PIN every time.”

scammers slid into his DMs. maya catches it before he hits pay.

a pre-payment check for UPI. it reads the scam’s stage, spots scammers testing the limits, and checks it’s really him tapping, all before the money leaves.

SELECTED CHALLENGE	02 • AI-driven scam pattern recognition
BUILT FOR	older and first-time big UPI payers; bank fraud analysts
STOPS	scam payments the victim approves, which can’t be undone

SAZI NIT Warangal

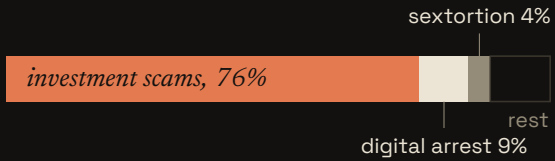


₹22,931 crore, gone in a year*

28 lakh
fraud complaints in 2025*

98.5%
of it went in transfers over ₹10,000*

WHERE IT WENT



Digital arrest got the headlines, and cases dropped 86% once people caught on*.

The money quietly moved to investment scams. Slower, sneakier, paid out over days.

Meet Ramesh. He's about to lose ₹75,000.

61, retired teacher, Warangal. A made-up but typical case; amounts are illustrative.



	day 1	day 3	day 4	day 6	day 8	day 10
TODAY	nothing	nothing	nothing	goes through	goes through	goes through
WITH MAYA	nothing to see	nothing to see	noted, silently	a nudge, 3 reasons	held 1 h, Priya asked	never sent

Priya is his daughter, the trusted person he picked at setup; a hold reaches her too. He loses the ₹5,000 sent after the nudge. The next ₹70,000 stays put.

WHAT AI ADDS ON THE SCAMMER'S SIDE

scripted chats
one AI runs dozens of "mentors"*

cloned voices
a voice note from "your son"

fake proof
AI-made profit screenshots

The obvious fix, what's out there, and what's new

THE OBVIOUS FIX



roughly what RBI has floated. can't tell rent from a scam.*

ALREADY OUT THERE	READS	DOES	FALSE ALARMS	TELLS YOU	YOUR DATA
MuleHunter.AI*	mule accounts	flags mules	bank-tuned	nothing	bank's own
DoT's FRI*	reported numbers	risk tags	known numbers	stock text	central
Android alert*	screen sharing	warns, waits	unknown callers	stock text	on phone
maya	history, mule graph, phone, chat, voice	nudge, hold, or ask Priya	two signals must agree; "how sure"	top 3 reasons, plain words	chat + voice stay on phone

what's new: maya reads the other three as inputs, then adds what none of them cover: the journey over days, the scam's stage, and reasons a person can act on.

The maths, in four lines

the rule: a hold needs two signals to agree, and one must be payment history or the receiving account. Chat, voice or phone alone never stops a payment.

1 SCORE

$$\operatorname{logit} p_t = \sum_k f_k(\mathbf{x}_t) + \operatorname{clip}\left(\mathbf{w}^\top \mathbf{u}_t,\ -\delta,\ \delta\right)$$

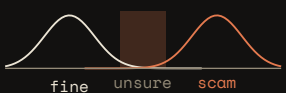
trees f_k read history, graph and probing. soft signals $\mathbf{u}_t$ (chat, voice, screen, typing) shift the odds by at most δ: they tip, never hold alone.



2 HOW SURE

$$\Gamma(\mathbf{x}_t) = \{y :\ 1 - \hat{p}_y(\mathbf{x}_t) \le \hat{q}_y\}, \quad \Pr\left[y_t \in \Gamma(\mathbf{x}_t)\right] \ge 1 - \alpha$$

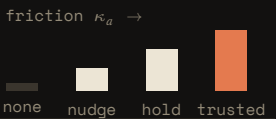
conformal prediction*: if "scam" and "fine" both fit, maya says it isn't sure and takes only a reversible step.



3 THE STEP

$$a^\star = \arg\max_a \ \left(p_t V_t e_a - \kappa_a - \lambda\, \mathbb{1}[a \neq \emptyset]\right)$$

money saved (V_t: payments still to come, e_a: how well step a works) minus annoyance κ_a; λ caps alerts at one a day.

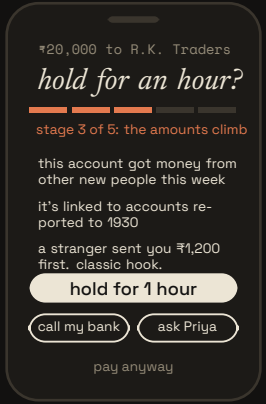


4 THE REASONS

$$\sum_k f_k(\mathbf{x}_t) = \phi_0 + \sum_j \phi_j(\mathbf{x}_t)$$

each feature's exact share of the score (tree SHAP*). the three biggest ϕ_j become the three lines on the alert.

What Ramesh sees on day 8



22% → *4%*

NO WARNING — TARGETED WARNING + CANCEL

of people paid a fraudster in a payments trial*

Long warning paragraphs barely helped. So: three short lines, a stage bar, two real buttons.

The lines are the model's own top three reasons (tree SHAP*) in plain words. The stage bar shows only if Ramesh shared the chat. Because he's on a call with his screen shared, the hold also goes to Priya.

How we'll test it

each part gets its own benchmark, its own metric, and a number to beat

PART	TESTED ON	MEASURED BY	HAS TO BEAT
payments + graph	IBM AML data*; our Hindi, Telugu, English journeys	recall at 1% false alarms; ₹ caught	*70% recall*
chat stage	PreScam, 11,573 real chats*; our own scripts	AUC; warning time before the money ask	*0.83 AUC*
voice note	ASVspoof 5*; open-licence Indic TTS	equal error rate (EER)	*AASIST's EER*
under attack	AI-reworded scripts, fresh mules, probing runs*	drop in catch rate after the attack	*under 10 points*
whole system	replayed journeys, later days only, 3 seeds	p95 latency; alerts per 100; coverage	*150 ms, 2, 90%*
fairness	slices by age, language, phone type	largest gap in miss rate	*3 points*

the number we'll lead with: ₹ of fraud stopped per 1,000 honest payments slowed, against RBI's flat one-hour rule.

Every test also holds out one whole scam type the model never trained on.

Safe by design, not by policy

CONSENT + PRIVACY

Opt in per signal (chat, voice, typing, screen); revoke any time. Only a label leaves the phone.

Never collected: UPI PIN, call audio, contacts, location, screen content, raw typing.

Banks learn together (FedAvg, DP, secure aggregation*); DPDP Rules*.

HUMAN REVIEW

Holds end by themselves in an hour; one alert a day at most.

Over ₹1 lakh, a bank analyst reviews before release.

Every hold is logged with the timeline, the reasons and the model version.

ATTACKERS + FAIRNESS

No scores shown. Probing (₹1, ₹2, ₹5 to one payee) is flagged.

Only bank-confirmed outcomes train it, so fake feedback can't poison it.

A miss-rate gap over 3 points by age, language or phone blocks a release.

WHEN MAYA IS WRONG, OR UNDER PRESSURE

it holds an honest payment	one tap releases it, or it ends in an hour
it misses a scam	bank rules still run; the confirmed case retrains it
the scammer says "ignore the bank"	Priya gets her own alert
maya breaks or drifts	the monthly check catches it; bank rules carry on

How a bank switches it on

shadow
scores real traffic silently; analysts see it, users don't

opt-in pilot
seniors and first-time big payers, one alert a day max

default on
for new-payee transfers, with the trusted-person path

shared
banks learn together without sharing data*

WHERE IT PLUGS IN

a small SDK in the bank's app
one scoring API in the bank's risk engine, called before UPI authorisation
no change to NPCI rails; fits RBI's 2025 rules that allow extra risk-based checks*
reads DoT's FRI and MuleHunter.AI; files 1930 / NCRP reports

WILL IT SCALE?

UPI handles about 66 crore payments a day*. Most never reach the model: known payees and small amounts take a fast path.

The model scores a payment in under a millisecond, so cost sits in lookups, not maths.

AWS path: Kinesis, DynamoDB, Neptune and GraphStorm on SageMaker*.

next: IMPS, wallets and AEPS; with a telco, the voice check moves onto the call.

The 48 hours

0–12 h	12–24 h	24–36 h	36–48 h
journey + graph features; probe counter; mock FRI feed	chat-stage model in 3 languages; voice check; screen + typing flags	fusion, “how sure”, action ladder; alert, stage bar, analyst screen	attack ourselves; fairness slices; latency test; demo

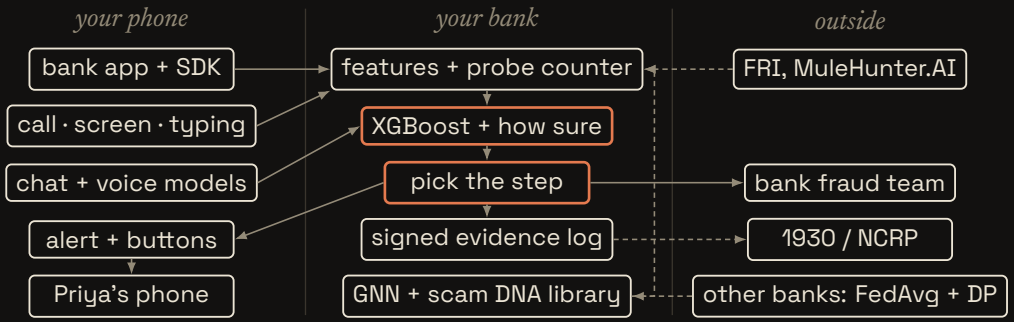
real in the demo: journey generator, XGBoost, “how sure”, SHAP reasons, alert screen (first run in appendix B4). mocked: FRI, MuleHunter.AI, other banks.

demo: ₹40k to mom goes through; an investment scam is held at payment 3; a fake-police call gets a hold and Priya; an unclear case gets a short hold he releases. Then a scam script it has never seen, reworded by AI, and miss rates by language and age.

the ask: one bank, shadow mode, one month of real traffic.

most people should never notice maya. the few who do should be glad she showed up.

Architecture 1 / 3: the whole system



SOLID ARROWS RUN BEFORE YOU HIT PAY. HOT PATH, p95 TARGETS:

`lookups ≤ 60 ms • model + policy ≤ 10 ms • network ≤ 80 ms • total ≤ 150 ms`

Only flags and labels leave the phone. Dashed arrows run in the background: the GNN refreshes account risk every few minutes and clusters new scam journeys into a "scam DNA" library; it replaces the simple graph features only if it wins on later months.

Architecture 2 / 3: the signals, in symbols

PAYMENT HISTORY, LAST TWO WEEKS

$n_t = \left|\{\text{new payees in } (t-7\text{d},\ t]\}\right|$
new people you paid this week

$e_t = a_t \,/\, \max\{a_s :\ s \text{ new},\ s \geq t-14\text{d}\}$
how fast the amounts climb

$g_t = \mathbb{1}[\text{stranger credit, last 7 d}]$
the "profit" hook

$z_t = \log a_t - \text{median}_{s<t} \log a_s$
how unusual this is for you

all from data before payment t. trained against small, realistic edits to amount and timing*.

THE MULE GRAPH



$d_R(v) = \min_{r \in R_t} \text{hops}(v, r),\ \ \alpha_{uv} \propto e^{-\lambda(t-\tau)}$

$h_v \leftarrow \sigma\left(W h_v + \sum_{(u \to v,\ \tau)} \alpha_{uv} M h_u\right)$

hops to a reported account; recent payments weigh more*

PROBING

$\pi_t = \left|\{s \in (t-1\text{h},\ t] :\ a_s \leq ₹10\}\right|$

tiny tries in the last hour: someone testing the limits before the real ask.

PHONE AND CHAT, CAPPED BY δ

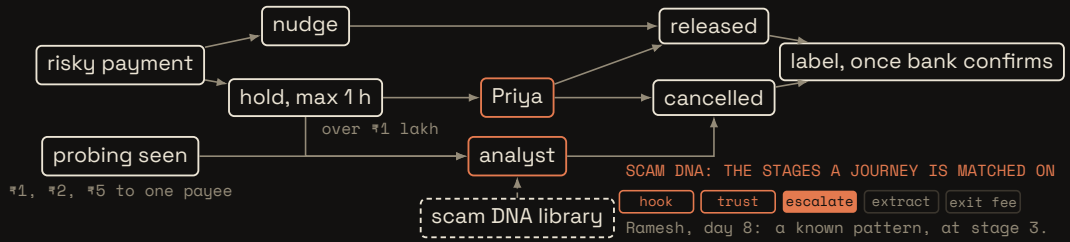
$c_t = \Pr[\text{stage}_t \geq 3 \mid \text{chat}]$
5 stages, hook to exit fee

$s_t = \mathbb{1}[\text{screen shared}]$

$v_t = \text{voice-spoof score}$

$b_t = \mathbb{1}[\|(\rho_t - \mu_u)/\sigma_u\|_\infty > 2]$
taps unlike yours. the PIN pad is NPCI's, never read.

Architecture 3 / 3: review and security



RISK	CONTROL
data leak from the bank	tokenised account IDs, field-level encryption, keys in an HSM
data shared between banks	FedAvg + DP noise + secure aggregation; server sees only the sum
insider misuse	two people to open an evidence record; every access logged
attacker learns the limits	no scores shown; probing flagged; retry limits; signed models
model drifts quietly	monthly “how sure” coverage check; fall back to rules only
poisoned feedback	only bank-confirmed outcomes are used to train

Appendix: where each criterion is answered

selected challenge: 02, AI-driven scam pattern recognition. Team SAZI, NIT Warangal.

CRITERION		SLIDES	EVIDENCE
problem fit	20	02, 03	MHA and NCRP figures; one user, one scam, day by day
technical architecture	20	05, A1–A3	four equations; 150 ms hot path; GNN as challenger
solution value	15	04, 06	vs RBI's delay, MuleHunter.AI, FRI, Android; 3-reason alert
feasibility	15	07, 10, B4	numbers to beat; 48-hour plan; a first run already done
safety and privacy	15	05, 08, A3	two-signal rule; chat + voice on phone; probing flagged
adoption and scale	10	09	SDK + one API; no NPCI change; fast path at 66 crore a day
communication	5	all	one story, Ramesh, 03 to 10; every number sourced

WHAT WE'RE ASSUMING, AND WHERE IT COULD BREAK

our simulator is easier than real traffic; every number on 07 is a target, not a result

scammers adapt, so we retest monthly and keep one scam type held out

if maya fails or drifts, the bank's own rules carry on alone

Android exposes call and screen-share state; iOS only reports screen capture*

the voice check reads shared voice notes, never live calls

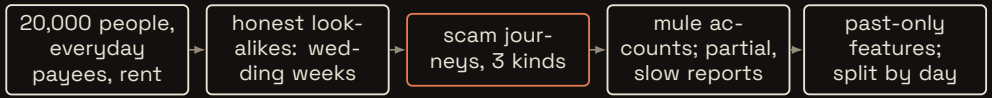
FRI and MuleHunter.AI are mocked until a bank gives access

Appendix: the research behind it

	THE FINDING		SO WE
1	a small model trained on real scam chats beat big chatbots at reading the scam's stage*	→	*run a tiny trained model on the phone*
2	AI rewording cut text detectors' catch rate by 88%; AI agents already run full scam calls*	→	*never let text block a payment alone*
3	simple graph features + XGBoost scored 62; the best time-aware graph network scored 76*	→	*ship the simple one; keep the GNN as challenger*
4	ordinary confidence scores break when patterns drift over time*	→	*recalibrate "how sure" monthly*
5	graph-attention models (AASIST) are a strong baseline for spotting synthetic speech*	→	*use it for voice notes; test on ASVspoof 5*

Appendix: how we make the data

no real bank data anywhere. everything below is synthetic or public.



PUBLIC DATA WE ADD IN THE FINALE

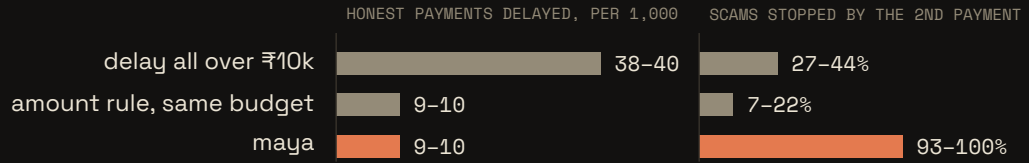
IBM synthetic AML transactions, for the graph model*
PreScam: 11,573 real scam chats with stage labels*
ASVspoof 5: real vs synthetic speech*
scam scripts we write in Hindi, Telugu, English and Hinglish; voices only from open-licence TTS with synthetic speakers, never a real person

HOW WE KEEP IT HONEST

test only on later days than training
hold out one whole scam type
3 random seeds; report the range, not the best
check amounts against public UPI and NCRP figures
say plainly that our own simulator is easier than reality

Appendix: a first run, end to end

on our own synthetic UPI simulator: 20,000 users, 30 days, 3 random seeds



4x fewer honest people slowed down, and most scams stopped before the second payment.

90–100%
of a scam type it never saw in training still caught

90–94%
"how sure" coverage, against a 90% target

0.4 ms
to score one payment (p95, model only, one CPU core)

Caveat: we wrote this simulator, and each test has about 30 scam journeys. It proves the pipeline, not real-world accuracy.

Appendix: models, code and assets we use

MODELS

XGBoost gradient-boosted trees (risk)

none of these see the UPI PIN; on-device signals are flags only

graph neural network in PyTorch Geometric (challenger)

a small open multilingual encoder, such as MuRIL or IndicBERT, fine-tuned (chat stage)

AASIST-style anti-spoofing model (voice note)

no large language model makes a decision or writes user-facing text

LIBRARIES

XGBoost (Apache-2.0)

scikit-learn, pandas, NumPy (BSD-3)

PyTorch (BSD-3), PyTorch Geometric (MIT)

Flutter (BSD-3), for the demo app

model weights and datasets under their own published licences, each checked before use

EVERYTHING ELSE

fonts: Newsreader, Space Grotesk, Space Mono (SIL Open Font Licence)

every diagram drawn by us in LaTeX/TikZ; no stock images or logos

no confidential bank, customer, NPCI, DoT or I4C data; FRI and MuleHunter.AI are mocked

AI assistant: Claude (Anthropic), for research, drafting and LaTeX; we checked every number against its source

Appendix: sources, 1 of 3

THE PROBLEM

* RBI, *Exploring safeguards in digital payments to curb frauds*, discussion paper, 9 Apr 2026.
* ThePrint, “Cybercrime saw 24% spike in 2025, Indians lost Rs 22,495 crore, mainly in investment scams”, Feb 2026 (MHA data).
* ThePrint, “After 465% spike in 2024, MHA data shows digital arrest scams are on a decline”, Feb 2026.
* Griffin & Mei, *How do crypto flows finance slavery? The economics of pig butchering*, SSRN 4742235, 2024.
* Acharya & Holz, *An explorative study of pig butchering scams*, arXiv:2412.15423, 2024.
* PIB, UPI completes 10 years, 30 Apr 2026 (24,162 crore payments in FY 2025–26).

THE ECOSYSTEM

* RBI, Authentication Mechanisms for Digital Payment Transactions Directions, 2025, in force 1 Apr 2026.
* DoT / PIB, Financial Fraud Risk Indicator and Digital Intelligence Platform, 12 Feb 2026.
* TechCrunch, “Google steps up AI scam protection in India”, 20 Nov 2025.
* MeitY, *Digital Personal Data Protection Rules, 2025*, notified 13 Nov 2025.
* Swift, federated learning trials with 13 banks, press release, 15 Sep 2025.
* AWS, *Modernize fraud prevention: GraphStorm v0.5 for real-time inference*, 2025.

THE MATHS

* Chen & Guestrin, *XGBoost*, KDD 2016.
* Vovk, Gammerman & Shafer, *Algorithmic Learning…*, 2005.
* Angelopoulos & Bates, arXiv:2107.07511, 2021.

Appendix: sources, 2 of 3

THE RESEARCH

* Akesson, Gathergood & Quispe-Torreblanca, *Preventing payments fraud in the FinTech era*, CeDEx DP 2023-08, Univ. of Nottingham.
* Sun, Ma, Li et al., *PreScam*, arXiv:2605.12243, 2026.
* Cheng, Sadasivan, Saberi, Saha & Feizi, *Adversarial paraphrasing*, arXiv:2506.07001, 2025.
* Badhe, *ScamAgents*, CAMLIS 2025, PMLR 299.
* Gounoue, Sao & Gottschalk, *TeMP-TraG*, arXiv:2503.16901, 2025.
* Blanuša et al., *Graph Feature Preprocessor*, ICAIF 2024.
* Wang, Kang, Yan, Kulkarni & Zhou, *Non-exchangeable conformal prediction for temporal GNNs*, KDD 2025.
* Chen, Gong, Cheng & Jin, *Temporal graph prototype-conditioned conformal prediction for fraud detection*, KDD 2026.
* Lundberg et al., *From local explanations to global understanding with explainable AI for trees*, Nature Machine Intelligence, 2020.
* Jung et al., *AASIST: audio anti-spoofing using integrated spectro-temporal graph attention networks*, ICASSP 2022.
* Wang et al., *ASVspoof 5*, 2024.
* Altman et al., *Realistic synthetic financial transactions for anti-money laundering models*, NeurIPS 2023.
* Ma et al., *PsyScam*, Findings of EMNLP 2025.

Appendix: sources, 3 of 3

PHONE SIGNALS

* Google, *Android expands pilot for in-call scam protection for financial apps*, 3 Dec 2025 (pilots in India and Brazil; warning and 30-second pause when a banking app opens during screen sharing on a call).
* Android Developers, screen recording detection (`addScreenRecordingCallback`), Android 15, 2024.
* Apple Developer, `UIScreen.isCaptured` / scene capture state.

ATTACKS ON THE MODEL

* Cartella et al., *Adversarial attacks for tabular data: application to fraud detection and imbalanced data*, SafeAI @ AAAI 2021, arXiv:2101.08030.

LEARNING ACROSS BANKS

* McMahan et al., *Communication-efficient learning of deep networks from decentralized data* (FedAvg), AISTATS 2017.
* Bonawitz et al., *Practical secure aggregation for privacy-preserving machine learning*, ACM CCS 2017.
* Abadi et al., *Deep learning with differential privacy*, ACM CCS 2016.